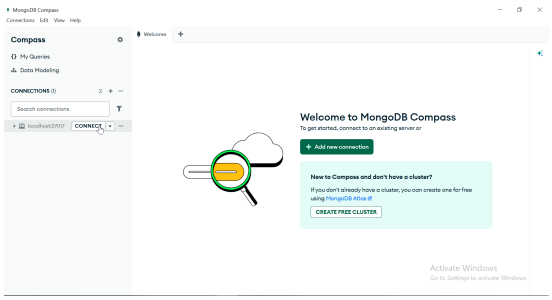
LAB 1

1 Set Up Data Base

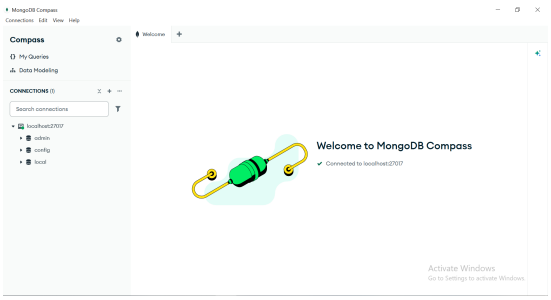
1.1 Connection

Connection

By default MongoDB creates a connection to the localHost at the port 27017 , if we downloaded the server MongoDB we should be able to connect just by clicking the connect button.



Click Connect

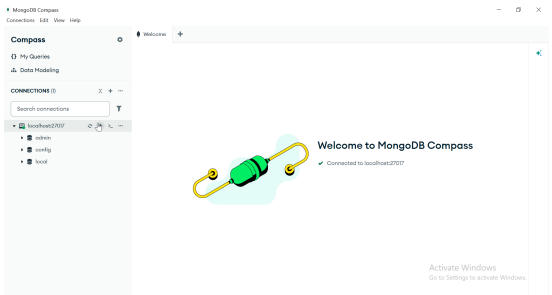


Connection Successful

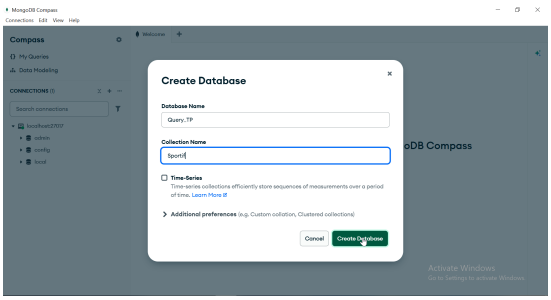
1.2 Create DataBase & Its Collections

Create

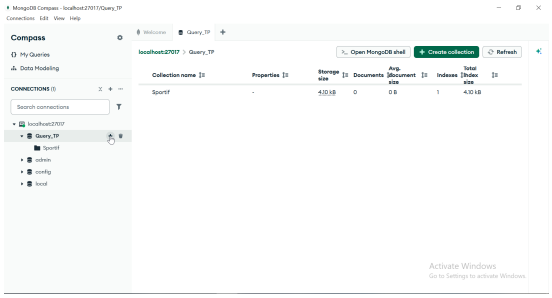
- We first create a new data base "Query_TP"
- Then we add 2 collections "Sportif" & "Gym"



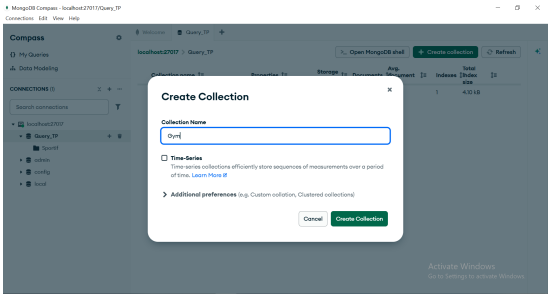
Click Create New Data Base



Input Data Base Name & Collection



Click Create New Collection

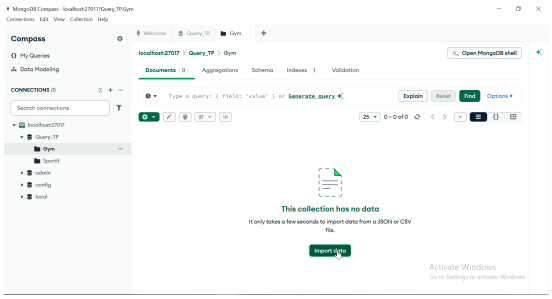


Input Name Of Second Collection

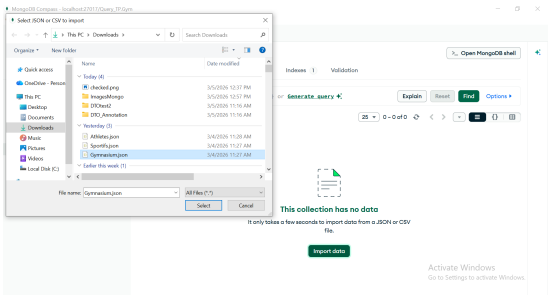
1.3 Import Json Document

Import

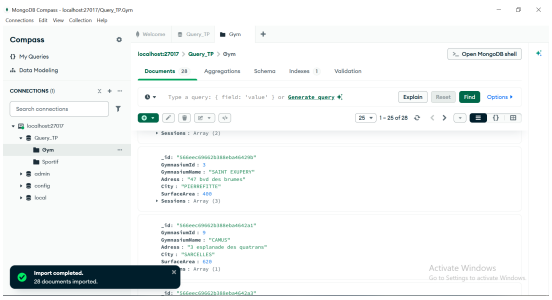
- Download the json files from the link https://drive.google.com/drive/u/0/folders/1QhHWxjCT5Tyiqe-nFsuYfgtquzwoyzJ4
- For each collection click the import data button then select the appropriate json file



Click On Import Data



Select Json Document

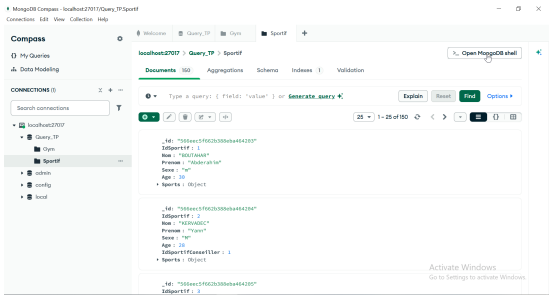


Import Completed

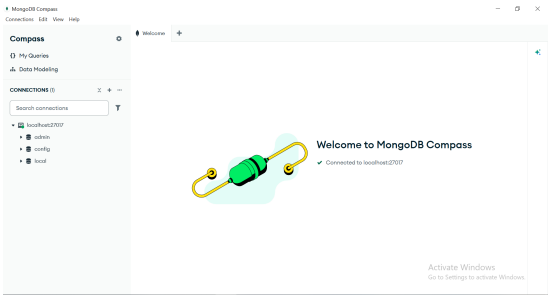
1.4 Open The Shell

Shell

- Click on the button open mongoDB shell
- Make sure to switch to the data base we created



Click On Open MongoDB Shell



Switch To The Data Base We Created

2 Queries

2.1 Sportifs That Are Between The Age Of 20 - 30

Mongo Query: Age 20-30

```
// first method implicit and
db["Sportif"].find({"Age" : {$gte : 20 , $lte : 30}} , {"Nom" : 1 , "Age" : 1, "_id": 0})

// second method explicit and
db["Sportif"].find( {
    $and : [
        {"Age" : {$gte : 20} } ,
        {"Age" : {$lte : 30} }
    ]
    } ,
    {"Nom" : 1 , "Age" : 1, "_id": 0}
)
```

2.2 Gyms From The Villetaneuse Or Sarcelles City & Surface Greater Than 400

Mongo Query: Gym In Villetaneuse Or Sarcelles , And Surface > 400

```
// first method using in operator
db["Gym"].find({
    "City" : {
        $in : [/SARCELLES/i, /VILLETANEUSE/i]
    },
    "SurfaceArea" : {$gt : 400}
})

// second method using or operator
db["Gym"].find({
    $or : [
        {"City" :/SARCELLES/i},
        {"City" :/VILLETANEUSE/i}
    ] ,
    "SurfaceArea" : {$gt : 400}
})
```

2.3 Sportifs That Practise Volley Ball

Mongo Query: Gyms Volley Ball

```
db["Sportif"].find( {"Sports.Jouer" : /Volley ball/i}, {"_id" : 0 , "Nom" : 1} )
```

2.4 Name Of Gyms & Day That Volley Ball Sessions

Mongo Query: Gyms Volley Ball

```
db["Gym"].find({"Sessions.Label" : /Volley ball/i } , {"Sessions.Day" : 1 , "GymnasiumName" : 1})
```

2.5 Same As Previous Query But Session Is On Wednesday After 15h

Mongo Query: Gyms Volley Ball Session On Wednesday After 15h

```
// first method implicit and
db["Gym"].find({
    "Sessions.Label" : "Volley ball"  ,
    "Sessions.Day" : /Mercredi/i ,
    "Sessions.Time" : {$gt : 15 }
    },
    {"_id" : 0, "GymnasiumName" : 1, "Sessions.Time" : 1}
)
```

Mongo Query: Gyms Volley Ball Session On Wednesday After 15h

```
// second method explicit and
db["Gym"].find({
    $and : [
    {"Sessions.Label" : "Volley ball"}  ,
    {"Sessions.Day" : /Mercredi/i} ,
    {"Sessions.Time" : {$gt : 15 }}
    ]
    },
    {"_id" : 0, "GymnasiumName" : 1, "Sessions.Time" : 1}
)
```

2.6 Sportifs That Practise Nothing

Mongo Query: Practise Nothing

```
db["Sportif"].find({
    $or : [
        {"Sports.Jouer" : {$exists : false} },
        {"Sports.Jouer" : {$size : 0} }
    ]
    },
    {"_id" : 0, "Nom" : 1 , "Prenom" : 1}
)
```